1. (6 marks) For independent, normal $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, provide the mean and variance of the following if they are necessarily normal. Give one line reason (see the last note in the instructions).

   A. $3X - 2$ B. $X + Y$ C. XY D. $|X|$ E. e^X F. $X - Y$

   Solution:

   - $\boldsymbol{3X - 2}$: **Normal**. Any affine transformation of a normal random variable is normal:
   $$3X - 2 \sim N(3\mu_X - 2, 9\sigma_X^2).$$
   - $\boldsymbol{X + Y}$: **Normal**. Since X and Y are independent normal random variables, their sum is normal:
   $$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$
   - $\boldsymbol{XY}$: **Not necessarily normal**. The product of two independent normal random variables is generally not normal. First, for a case when it is normal, take X unit normal and Y=constant.

     For the intuition behind the counter example, take X,Y unit normal. For small values of X, the product is getting pushed to the origin. For occasionally large values of X, the product becomes very large. The distribution concentrates on the zero, and the "tails" are very large. This is not the behaviour of a normal where the tails drop off very quickly;(s.d. of 3 or above are hard to find.)

     Teaching-team notes. This is good enough for the exam – noting that the question as a whole is worth only 6 marks. (Sahil Deo's piazza post is cleaner.)

     There is actually a formal argument for the tail behaviour. You can do this by (a) noting that MGF of a unit normal is defined at every value of t. MGF of XY is undefined for mod(t) > 1.

     The constant case is in fact one of the rare cases when the product is normal, and again, here we are skirting on the boundary situation.
   - $\boldsymbol{|X|}$: **Not normal**. For example, if $X \sim N(0, 1)$, then $|X|$ has a half-normal distribution, not a normal distribution. This is not normal as it doesn't take negative values.
   - $\boldsymbol{e^X}$: **Not normal**. If X is normal, then e^X has a log-normal distribution. This is not normal as it doesn't take negative values.
   - $\boldsymbol{X - Y}$: **Normal**. Since $-Y$ is normal and independent of X, their sum $X + (-Y) = X - Y$ is normal:
   $$X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

   Hence, the transformations that are necessarily normal are

   $$\boxed{3X - 2, \quad X + Y, \quad X - Y}.$$